Who does what

Four people, four lanes, one pipeline. Each person owns a set of modules and the file that crosses their boundary — so after week 2, nobody waits on anybody until real data lands.

A — PERSON A — Data & Corpus

NAME

Gets real data in, deduplicated, split without leakage, and frozen.

PICK	Your most careful person. Bugs here are invisible and fatal.

OWNS	`router/data.py`
	`router/dedup.py`
	`router/splits.py`
	`scripts/01_acquire_routerbench.py`
	`configs/datasets.yaml`

HANDS OFF	`corpus.parquet` · `scores.parquet`
CHAPTER	Datasets, deduplication and split methodology

B — PERSON B — Features & Router

NAME

Turns prompts into features, and features into two calibrated predictions.

PICK	Whoever wants the most classic hands-on ML work.

OWNS	`router/features.py`
	`router/models.py`
	`router/calibrate.py`
	`router/graders.py`

HANDS OFF	`features.parquet` · `predictions.parquet`
CHAPTER	Feature engineering and router architecture

C — PERSON C — Evaluation & Statistics

NAME

Turns predictions into a Pareto curve and a threshold τ^* you can defend.

PICK	Your strongest at statistics, and the most willing to write tests.

OWNS	`router/costs.py`
	`router/eval.py`
	`router/threshold.py`
	`router/synthetic.py`
	`plots/plot_pareto.py`
	`scripts/estimate_spend.py`

HANDS OFF	the Pareto figure · the ablation table
CHAPTER	Cost modelling, threshold selection and distribution-free guarantees

D — PERSON D — Systems & Demo

NAME

Makes every model callable, every run reproducible, and the demo live.

PICK	Whoever is happiest with infrastructure and debugging.

OWNS	`router/generate.py`
	`router/serve.py`
	`.github/workflows/`
	`manifest tooling`
	`requirements*.txt`

HANDS OFF	`responses.parquet` · the inference layer
CHAPTER	Local inference, serving architecture and the live prototype

PAGE 2 the handoff map — who sends what to whom, and when

PAGE 3 the 16 weeks, the five gates, the weekly rhythm

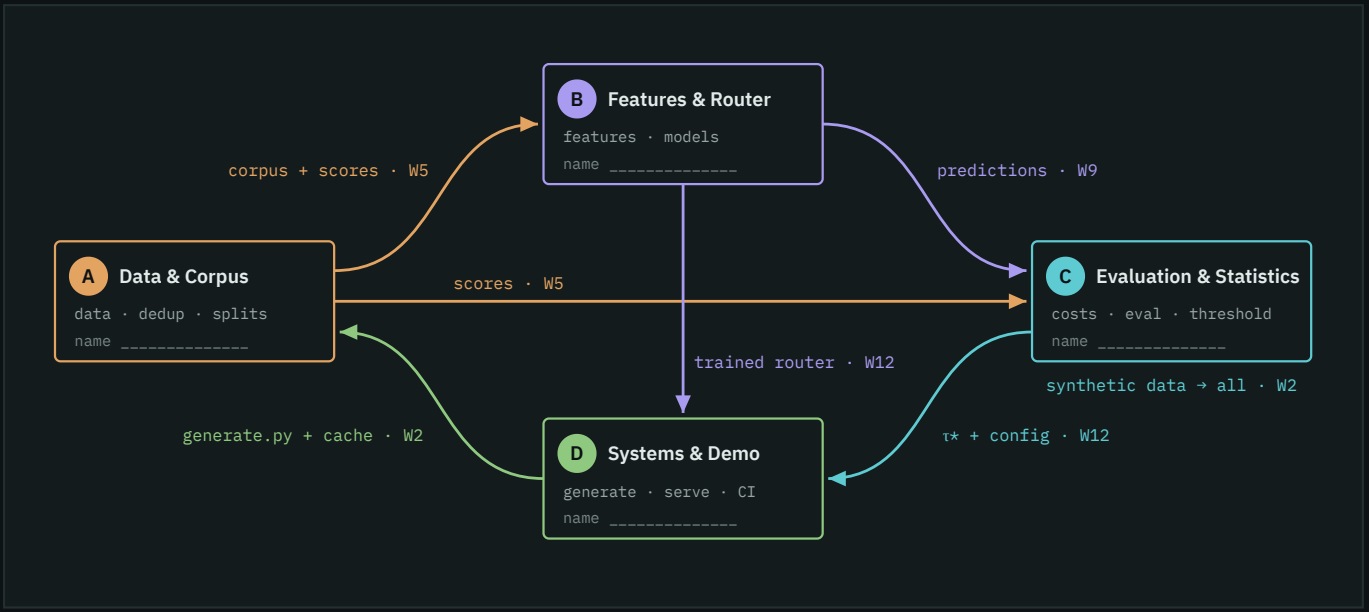
PAGES 4–7 one page per person, week by week

PAGE 8 the responsibility matrix and the team rules

THE MAP

Every handoff, and the week it's due

Arrows are files that cross from one person to another, coloured by who sends them. If your arrow is late, the person at the other end is blocked — so these dates matter more than any internal deadline.

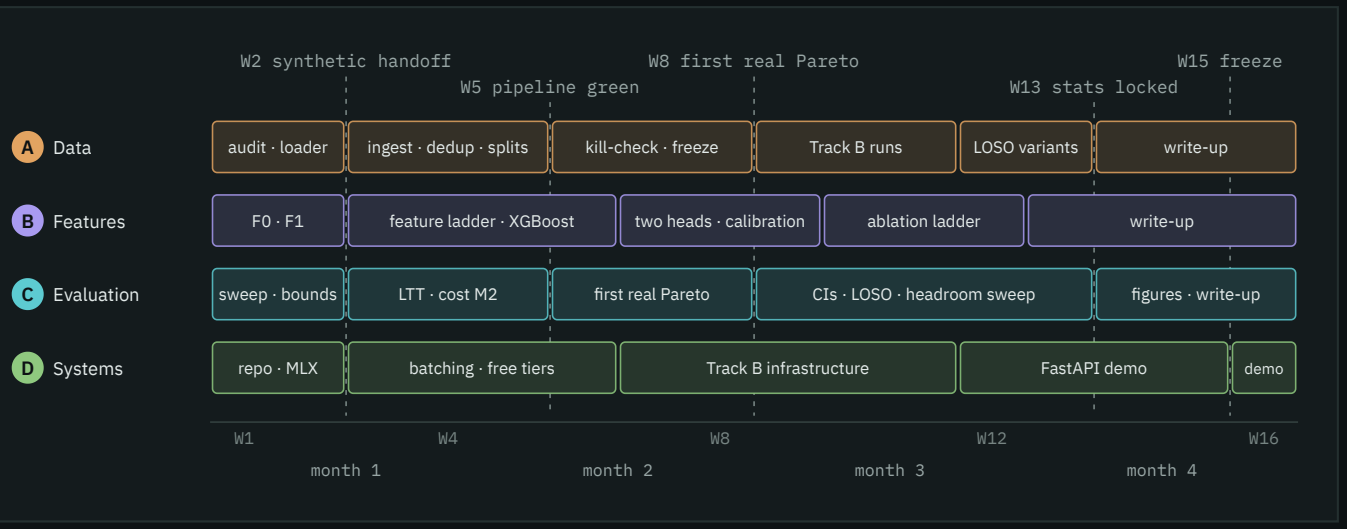


WEEK	FROM → TO	WHAT	WHY IT MATTERS
W2	C → all	synthetic `scores.parquet` — already built	B, C and D build and test before any real data exists
W2	D → A	`generate.py` + cache	A runs every Track B job through it from week 9
W2	D → C	measured tok/s in `models.yaml`	Replaces every estimated hour figure in the design
W5	A → B, C	first real `corpus` + `scores`	B trains on real features; C runs the real sweep
W8	A → all	frozen corpus + manifest v1	Nothing is regenerated after this
W9	B → C	`predictions.parquet`	The input to τ selection
W11	A + D → C	Track B scores	The headroom-sweep result
W12	B + C → D	trained router + τ*	What the demo actually runs
W13	C → all	locked numbers + figures	No chapter's numbers change after this

THE SCHEDULE

Sixteen weeks, four lanes in parallel

After the two-week foundation, all four lanes run at once. B, C and D build against synthetic data until A's real corpus lands in week 5 and freezes in week 8.



THE FIVE GATES — WHAT "DONE" MEANS AT EACH DASHED LINE

WEEK	GATE	SIGNS OFF	EVIDENCE
W2	**Contracts frozen**	all four	Schemas agreed in `schema.py`; measured tok/s committed; one merged PR per lane.
W5	**Pipeline green on synthetic**	C	End-to-end run on fake data draws a Pareto curve. Plumbing proven before real data.
W8	**First Pareto on real data**	A + C	Pair chosen by headroom; kill-check passed. **If it slips, cut scope, not rigour.**
W13	**Statistics locked**	C	CPT(95%) + savings on the untouched test split, with CIs. No number changes after.
W15	**Freeze**	all four	Demo running, figures final, draft done. Week 16 is buffer and rehearsal.

THE WEEKLY RHYTHM

MON · WED · FRI

Async standup in the group chat, three lines each: *done · next · blocked*.

FRIDAY, 45 MIN

Integration. One person merges open PRs and runs everything end to end.

INTEGRATOR ROTATION

D W1 → A W2 → B W3 → C W4, then repeat.

A

PERSON A

Data & Corpus

NAME _____

Get real data in — deduplicated, split without leakage, and frozen — so every number downstream stands on ground that doesn't move.

OWNS

`router/data.py`
`router/dedup.py`
`router/splits.py`
`scripts/01_acquire_routerbench.py`
`configs/datasets.yaml`

RECEIVES

`W2` from **D** — generate.py + cache, for Track B later

`W2` from **C** — synthetic data with planted duplicates, to test dedup

HANDS OFF

`W5` to **B**, **C** — first real corpus.parquet + scores.parquet

`W8` to all — frozen corpus + manifest v1

`W11` to **C** — Track B scores, one file per small model

WEEK BY WEEK

WEEKS	WORK	DONE WHEN
1	**RouterBench audit** — answer the five questions in `notes/routerbench_audit.md`. Write `load_routerbench()` mapped onto the CORPUS and SCORES contracts; record which model is S and which is L in `datasets.yaml`.	A 200-row sample passes both contracts in `tests/test_data.py`
2	**Dedup** — MinHash on 5-gram shingles (Jaccard ≥ 0.8) plus embedding cosine (≥ 0.95), connected components → `group_id`. Add leave-one-subject-out splits.	≥ 0.9 recall on the 100 planted duplicate pairs
3–5	**Full ingest** of GSM8K, MMLU, MT-Bench and MBPP; token counts for re-pricing; grouped 60/20/20 splits; dedup report.	`assert_no_leakage` passes; parquets handed to B and C
6–8	**Headroom kill-check** for each candidate S/L pair — keep the pair with the largest $Q_L - Q_S$. Hand-check 50 items per grader with B. Freeze the corpus.	Headroom ≥ 0.05 on every dataset kept; manifest v1 on the drive
9–11	**Track B** — run Llama 3.2 3B, Qwen3.5-4B and Phi-4-mini overnight on GSM8K + MMLU using D's infrastructure; align items to RouterBench's frontier scores.	Three scores files handed to C
12–13	Leave-one-subject-out and nested-CV variants on request from C; dataset tables for the report.	C can re-run any split with one flag
14–16	Write your chapter.	Draft complete

WATCH OUT FOR

- A near-duplicate across train/test inflates every number, and nothing downstream catches it. Run `assert_no_leakage` after every split.
- Never re-download or regenerate without bumping the manifest version.
- A fanless M2 Air throttles 15–25% under long loads. Run Track B overnight, plugged in.

YOUR CHAPTER

Datasets, deduplication and split methodology

YOU REVIEW

B's PRs

B

PERSON B

Features & Router

NAME _____

Turn prompts into features, and features into two calibrated predictions: how well the small model will do on this query, and how well the frontier model will.

OWNS

`router/features.py`
`router/models.py`
`router/calibrate.py`
`router/graders.py`

RECEIVES

`W2` from **C** — synthetic corpus + scores, to build against

`W5` from **A** — real corpus.parquet + scores.parquet

HANDS OFF

`W9` to **C** — predictions.parquet from the calibrated two-head model

`W12` to **D** — the trained router, exported for the demo

WEEK BY WEEK

WEEKS	WORK	DONE WHEN
1	**F0**: the ~25 heuristic features as a pure function. **F1**: TF-IDF, word 1–2 grams + char 3–5 grams, fit on the train split only.	Hand-written prompts give hand-computed F0 values; token count ρ > 0.3 with synthetic difficulty
2	**F2**: `bge-base-en-v1.5` embeddings cached to `features.parquet`. Logistic-regression baselines on F0 and on F1.	Spearman ρ($\hat{\Delta}$, Δ) reported for each; F1 ≥ F0
3–6	`graders.py` (numeric exact match, MCQ logprob). XGBoost on fused F0 + F2. Move the whole ladder onto real data when A hands off at W5.	Every feature set has a validation ρ on real data
7–9	**Two-head MLP** (793 → 256 → 2), BCE + gap loss, 5 seeds. Isotonic calibration per head; reliability diagrams and ECE.	`predictions.parquet` handed to C
10–12	**Ablation ladder**: F0, F1, XGBoost, F2, F2 + F0, cost-normalized rule v2, F4 logprob cascade. F3 fine-tune overnight on MPS if time allows.	Every ablation row has numbers; router exported for D
13–16	Write your chapter.	Draft complete

WATCH OUT FOR

- Fit TF-IDF, scalers and calibrators on train (or cal) only. Fitting on everything is leakage.
- Select models by ranking quality — Spearman ρ — not accuracy. The threshold sweep consumes the ranking.
- Never tune on synthetic data. It tests plumbing, not hyperparameters.

YOUR CHAPTER

Feature engineering and router architecture

YOU REVIEW

C's PRs

C

PERSON C

Evaluation & Statistics

NAME _____

Turn predictions into a Pareto curve and a threshold τ* you can defend in front of an examiner. This lane is the project's actual contribution — staff it first.

OWNS

`router/costs.py`
`router/eval.py`
`router/threshold.py`
`router/synthetic.py`
`plots/plot_pareto.py`
`scripts/estimate_spend.py`

RECEIVES

`W2` from **D** — measured tok/s for configs/models.yaml

`W5` from **A** — real scores.parquet

`W9` from **B** — predictions.parquet

`W11` from **A** + **D** — Track B scores

HANDS OFF

`W2` to all — synthetic data (already built)

`W12` to **D** — τ* and the router config for the demo

`W13` to all — locked numbers and final figures

WEEK BY WEEK

WEEKS	WORK	DONE WHEN
1	`eval.py` : Q(τ), C(τ), PGR, APGR, random and oracle curves. `threshold.py` : the exact O(n log n) sweep and the concave envelope. Work a 6-row example on paper.	Property test: the sweep matches brute force on 200 random cases
2	**Lower bounds** — Hoeffding and paired bootstrap; τ* selection; `plot_pareto` panels A and B. Stress-test on low-headroom synthetic data.	First (synthetic) Pareto figure with random, oracle and τ*
3–5	**Learn-then-Test** fixed-sequence selection. Cost model M2 and the local-vs-Haiku crossover. Panels C and D.	Whole pipeline green on synthetic — W5 gate
6–8	First sweep on A's real scores; headroom numbers for the kill-check.	First real Pareto figure — W8 gate
9–13	Paired-bootstrap CIs everywhere; the LTT bound; grouped vs leave-one-subject-out; the headroom sweep (CPT(95%) vs $Q_L - Q_S$); savings vs a batch-optimized baseline.	CPT(95%) and savings on the untouched test split, with CIs — W13 lock
14–16	Final figures, the ablation table, and your chapter.	Draft complete

WATCH OUT FOR

- τ is chosen on `cal` only. The test split is opened once, at week 13.
- A τ-solver bug produces plausible, publishable, false numbers. Your tests matter most in the repo.
- Oracle PGR above 1.0 is correct, not a bug: sometimes the small model beats the frontier.

YOUR CHAPTER

Cost modelling, threshold selection and distribution-free guarantees

YOU REVIEW

B's PRs, and every schema change

PERSON D

D Systems & Demo

NAME _____

Make every model callable through one interface, make every run reproducible, and make the router something you can show working live at the defence.

OWNS

```
router/generate.py
router/serve.py
.github/workflows/
manifest tooling
requirements*.txt
```

RECEIVES

- `W12` from **B** — the trained router
- `W12` from **C** — τ* and the router config

HANDS OFF

- `W2` to **A** — generate.py + content-hash cache
- `W2` to **C** — measured tok/s in configs/models.yaml
- `W9–11` to **A** — overnight run infrastructure for Track B

WEEK BY WEEK

WEEKS	WORK	DONE WHEN
1	GitHub repo, CI, branch protection, PR template. **Benchmark MLX** — Qwen3.5-4B and Llama 3.2 3B at batch 1 and 8. `generate.py` : one Client protocol, `MLXClient` + `OpenAICompatClient` , content-hash cache.	A repeated request hits the cache; measured tok/s committed
2	End to end on 20 real GSM8K items — local MLX and one free-tier API, both through the cache.	Re-running makes zero model calls
3–6	Batched MLX decoding. Gemini, GitHub Models and Groq clients with rate-limit backoff and retry. Manifest + shared-drive sync script.	Batch-8 throughput measured; a 200-item run survives kill + resume
7–11	Track B overnight-run infrastructure with A: resumable, logged, thermal-aware.	Three small-model runs complete
12–15	**FastAPI demo** `/route` : features → router → dispatch to local or API; telemetry (routed_to, estimated savings, latency); live cost counter.	Router p99 < 15 ms on CPU
16	Defence rehearsal. Record a backup video of the demo.	Backup video exists

WATCH OUT FOR

- API keys live in `.env` only. A committed key is a revoked key.
- Free tiers change limits without notice. The cache is what saves you.
- Live demos fail at defences. Record the backup video.

YOUR CHAPTER

Local inference, serving architecture and the live prototype

YOU REVIEW

A's PRs

THE MATRIX

One owner per job. Always.

Owns means does the work and answers for it. **Reviews** means must approve the PR before it merges. **Input** means supplies something the owner needs. Every row has exactly one owner — if two people think they own something, nobody does.

JOB	A	B	C	D
RouterBench ingest + model mapping	OWNS		REVIEWS	
Dedup + grouped / LOSO splits	OWNS		REVIEWS	
Graders + 50-item hand checks	REVIEWS	OWNS		
Feature ladder F0–F2 + XGBoost		OWNS	REVIEWS	
Two-head router + calibration		OWNS	REVIEWS	
Price table + cost model			OWNS	INPUT
τ sweep, bounds, Learn-then-Test		REVIEWS	OWNS	
Pareto figures + ablation table		INPUT	OWNS	
Model clients + cache	REVIEWS			OWNS
Track B overnight runs	OWNS			INPUT
Manifests + reproducibility	REVIEWS			OWNS
FastAPI demo		INPUT	INPUT	OWNS
Contract changes (schema.py)	REVIEWS	REVIEWS	REVIEWS	REVIEWS
Friday integration	FROM W2	FROM W3	FROM W4	FROM W1

FIVE TEAM RULES

1

PRs are reviewed from another lane. By the defence, all four of you can explain every stage.

2

Nobody regenerates responses without bumping the manifest version.

3

Artifacts are not git. Shared drive, named by content hash, never emailed.

4

No paid API calls before week 8. Free tiers only.

5

Every owner writes their own tests — and C's matter most.

Fill in the four names on page 1 and on pages 4–7, then pin this PDF in the group chat.